Dr. Prathmesh Bhadane
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Education

Degree	Institution	CPI / %	Year
PhD (Materials Engineering)	Indian Institute of Technology, Gandhinagar	8.84/10	2021-2026
MTech (Material Science and Engineering)	Indian Institute of Technology, Gandhinagar	8.00/10	2019-2021
BE (Mechanical Engineering)	Mumbai University	9.18/10	2014-2018

- **Award:** Commendation for **Outstanding Innovation (PhD)** by Indian Institute of Technology, Gandhinagar during the Awards Ceremony of the 15th Convocation.

Post-doctoral Fellow, Chemistry, IIT Bombay (09-Feb-2026 to Present)

- Working on designing Metal-organic frameworks (MOFs) based heterogeneous catalysts for gas conversion (CO_2, CH_4) into useful chemicals under the guidance of Prof. Deepti Kalsi (Indian Institute of Technology, Bombay).
- Industry Project- Heterogeneous Catalysts for Selective Ethylene Oligomerization toward Sustainable Aviation Fuel (SAF) Precursors.
- Teaching assistance for course CH-831L- Instrumentation laboratory, Chemistry at IIT Bombay.

Visiting PhD scholar, University of Birmingham, United Kingdom (04-Sep-2024 to 15 Jan-2025)

- Worked under the joint supervision of Dr. Swaroop Chakraborty and Prof. Iseult Lynch (School of Geography, Earth & Environmental Science, University of Birmingham, UK) on MOF-based environmental remediation and resource recovery.
- Conducted research on Safe and Sustainable by Design MOF Beads for Selective Entrapment of Rare Earth Elements, developing MOF-polymer composite beads for sustainable REE recovery.
- Outcomes: Two Research articles and one patent: (1) Bhadane P; et al, Environmental Science: Nano (2025), (2) Bhadane P; et. al, Environmental Science & Technology 59 (2025): 16379–16391 and UK Patent Application Number 2611413.2.

PhD research work

- **Research Focus:** Sustainable Nanocomposites and Metal-Organic Framework (MOF) Systems for Environmental Remediation and Resource Recovery.
- **Projects-** (1) Tap water filtration system using antibacterial polymer coated beads, (2) Adsorptive removal of heavy metal and adsorption of REEs using MOFs, (3) Natural fibers and graphene oxide reinforced bio-composite materials, (4) Sustainable building materials from industrial waste by-products of coal-fired thermal power plants.
- **Advisors & International Collaborators:** Worked under the supervision of Prof. Abhijit Mishra (IIT Gandhinagar, India), Prof. Superb Misra (IIT Gandhinagar, India), Prof. Iseult Lynch (University of Birmingham, UK) and Prof. David Fairen-Jimenez (University of Cambridge, UK).

Teaching and Mentoring Experience

- Assistant Teacher at Shifa's Group Tuitions, Ambarnath, Thane [August 2014-February 2018]
  - Taught Mathematics to students of classes XI and XII, ensuring conceptual clarity and problem-solving proficiency.
- Private Tutor – Mechanical Engineering Subjects [July 2016-May 2018]
  - Conducted short courses on core Mechanical Engineering subjects, including Strength of Materials, Heat Transfer, Fluid Mechanics, Machine Design, for a batch of 40 undergraduate students.
- Independent Instructor, Materials Characterization Techniques, IIT Gandhinagar [January–May 2024]
- Research Supervision:
  - Guided Mr. Adit Rambhia, a final-year B.Tech student at IIT Gandhinagar, on a one-year research project titled "Development of a Reusable Semi-IPN Polymer-Coated Cellulose Acetate Bead Filtration System for Real-World Antibacterial Applications" (2024-2025).
  - Guiding Mr. Sukanta Bar, MSc student at IIT Bombay, on project titles "MOF derived heterogeneous catalysts for methane oxidation" (April 2026-Ongoing).

List of Peer-reviewed publications and attended conferences

Peer-reviewed research papers

1. **Prathmesh Bhadane,** R. Gayathri, M. Dotiyal, A. Chootray, S. Misra, D. Kalsi, S. Chakraborty "Adsorption-Programmed Transformation of a Two-Dimensional Cu–Imidazolate Framework into Mn-Integrated Catalytic Nanonetwork." Small (2026): e75834. **IF: 11.8** DOI: https://doi.org/10.1002/smll.75834

2. **Prathmesh Bhadane**, D. Menon, P. Goyal, M. R. A. Kiapi, B. K. Satpathy, A. Lanza, I. Mikulska, R. Scatena, S. Michalik, P. Mahato, M. Asgari, X. Chen, S. Chakraborty, A. Mishra, I. Lynch, David Fairen-Jimenez, Superb K. Misra, "A Two-Dimensional Metal-Organic framework for efficient recovery of heavy and light rare earth elements from electronic wastes. Separation and Purification Technology (2024), 130946". **IF: 9.1** DOI: https://doi.org/10.1016/j.seppur.2024.130946

3. **Prathmesh Bhadane**, Pankti Dhumal, Emilie Brun, Andrew Britton, Iseult Lynch, and Swaroop Chakraborty. "Safe and Sustainable by Design MOF Beads for Selective Entrapment and Recovery of Rare Earth Elements." Environmental Science & Technology (2025). **IF: 12.2** DOI: https://doi.org/10.1021/acs.est.5c03112

4. **Prathmesh Bhadane** and Swaroop Chakraborty, "Cross-Material Synergies of Carbon Nanomaterials, MOFs, and COFs: Innovative Approaches for Sustainable Environmental Remediation and Resource Recovery" Coordination Chemistry Reviews (2025). **IF: 25.6** DOI: https://doi.org/10.1016/j.ccr.2025.216669

5. Dhumal Pankti*, **Prathmesh Bhadane***, Bashiru Ibrahim, and Swaroop Chakraborty. "Evaluating the Path to Sustainability: SWOT Analysis of Safe and Sustainable by Design Approaches for Metal-Organic Frameworks." Green Chemistry (2025). (***equal contribution**). **IF: 10.6** DOI: https://doi.org/10.1039/D5GC00424A

6. **Nature Protocols (2026):** Dhruv Menon, **Prathmesh Bhadane**, Prateek Goyal, Priya Mahato, Swaroop Chakraborty, Iseult Lynch, Superb Mishra**, "**An end-to-end workflow for selective heavy metal capture and rare-earth element recovery using metal–organic frameworks". **IF:16.0 DOI:** https://doi.org/10.1038/s41596-026-01425-y

7. **Prathmesh Bhadane**, Deepti Kalsi, Rethinking MOFs for REE recovery: Role of functional groups, selectivity limitations and possible future directions", Coordination chemistry Reviews (2026) **IF: 25.6** DOI: https://doi.org/10.1016/j.ccr.2026.218197

8. **Prathmesh Bhadane**, Deepak G. Prajapati, Adit Rambhia, Mayank Dotiyal, Pramina Kumari Pandey, Deepak Rajput, and Abhijit Mishra. "Antibacterial Activity of Cellulose Acetate–Graphene Oxide Composites: A Comprehensive Multimethod Assessment." ACS Omega (2025). **IF:5.2** DOI: https://doi.org/10.1021/acsomega.5c04982

9. **Prathmesh Bhadane**, P. Mahato, D. Menon, B. K. Satpathy, L. Wu, S. Chakraborty, P. Goyal, I. Lynch, and Superb K. Misra. "Hydrolytically stable nanosheets of Cu–imidazolate MOF for selective trapping and simultaneous removal of multiple heavy metal ions." Environmental Science: Nano (2024). **IF: 5.6** DOI: https://doi.org/10.1039/D3EN00754E

10. **Prathmesh Bhadane**, and Swaroop Chakraborty. "Cellulose Acetate-nanoMOF Beads: A Safe, Sustainable and Scalable Solution for Lead Remediation in Complex Water Systems." Environmental Science: Nano (2025). **IF: 5.6** DOI: https://doi.org/10.1039/D5EN00056D

11. **Prathmesh Bhadane,** and Abhijit Mishra. "The effect of Alkali treatment on Pineapple Leaf Fibers (PALF) on the performance of PALF reinforced rice starch biocomposites." Journal of Natural Fibers (2022): 1-15. **IF: 3.1** DOI: https://doi.org/10.1080/15440478.2022.2060406

12. **Prathmesh Bhadane**, and Abhijit Mishra. "Versatile, flexible rice starch–graphene oxide bio-nanocomposites." Environmental Science: Water Research & Technology (2024). **IF:3.5** DOI: https://doi.org/10.1039/D4EW00419A

13. Chakraborty, Swaroop, **Prathmesh Bhadane**, Wilgner Lima da Silva, Hiroto Kitaguchi, Ana Guilherme Buzanich, Sang Pham, Superb K. Misra, and Iuliia Mikulska. "Freeze-drying enables resource-efficient isolation of copper imidazolate metal–organic framework nanosheets for transformation-aware lead capture." Green Chemistry (2026). **IF: 10.6** DOI: https://doi.org/10.1039/d6gc03068h

14. Swaroop Chakraborty, Drexel Roland, **Prathmesh Bhadane,** Langford Nathan, Dhumal Pankti, Meier Florian, Lynch Iseult, "Integrated Multimethod Approach for Size-Specific Assessment of Potentially Toxic Element Adsorption onto Micro- and Nanoplastics: Implications for Environmental Risk" Nanoscale (2025). **IF: 5.1** DOI: https://doi.org/10.1039/D5NR00353A

15. Chakraborty, Swaroop, Iuliia Mikulska, Rhiannon Boseley, Sang Pham, **Prathmesh Bhadane**, Pankti Dhumal, Santanu Majumder et al. "Hierarchical Environmental Exposure Transforms Zeolitic Imidazolate Framework-8 and Increases Toxicity in Daphnia magna." ACS nano (2026). **IF: 17.3** DOI: https://doi.org/10.1021/acsnano.6c01107

16. Chakraborty, S., Guilherme Buzanich, A., **Prathmesh Bhadane**, Kitaguchi, H., Pham, S., & Mikulska, I. (2026). Structural Persistence Masks Commit-Point Chemical Transformation in Copper–Imidazolate Nanosheet Metal–Organic Frameworks. ACS nano (2026). **IF: 17.3** DOI: https://doi.org/10.1021/acsnano.6c06011

17. Saran, Rohitash, Sanchit Saxena, Hritaban Acharya, **Prathmesh Bhadane**, and Kaling Taki. "Utilization of Water Treatment Plant Sludge for Creating Green Bricks and Examining Its Gamma Radiation Shielding Potential." Journal of Hazardous, Toxic, and Radioactive Waste 29, no. 2 (2025): 04025009. **IF: 2.2**, DOI: https://doi.org/10.1061/JHTRBP.HZENG-1406

18. **Prathmesh Bhadane**, and Mishra Abhijit. "Preparation and characterization of graphene oxide and pineapple leaf fibers reinforced rice starch bio-composite." Materials Today: Proceedings (2022). **IF:1.9** DOI: https://doi.org/10.1016/j.matpr.2022.06.092

Selected Conference Presentations

1. **Prathmesh Bhadane**, Abhijit Mishra. "Effect of alkaline treatment of pineapple leaf fibers on the PALF reinforced rice starch composite" International Conference on Natural Fibers 2021, Portugal.
2. **Prathmesh Bhadane**, Abhijit Mishra, "Characterization of pineapple leaf fibers and graphene oxide reinforced rice starch bio composite" International Conference on Novel Materials and Technologies for Energy and Environmental Applications, 2022, India.
3. **Prathmesh Bhadane**, Abhijit Mishra, "Multifunctional Flexible Starch-Graphene Oxide Bio-Composite" MRSI AGM 2023, IIT BHU and EMRS Spring 2024, Strasbourg, France.
4. **Prathmesh Bhadane,** Swaroop Chakraborty, "MOF–polymer composite beads for water remediation" UoB research showcase, 2024, United Kingdom.
5. **Prathmesh Bhadane**, Deepti Kalsi, "Critical Mineral Recovery: If not now? Then when?" 3 Minutes Presentation at In-house symposium 2026, Chemistry, IIT Bombay

Patent

- **Prathmesh Bhadane**, Goyal Prateek, Mahato Priya and Misra Superb K., "Adsorbent and a process for recovery of rare Earth elements", Indian Patent Office (IPO), Patent Application No.: 202421047907, Jun. 21, 2024.
- Swaroop Chakraborty, **Prathmesh Bhadhane**, UK Patent Application Number 2611413.2 - Metal Organic Framework Particles, May 18, 2026

Professional Certificates

- Certificate of completion in Professional in Product Design and Analysis from CADD center training services, Mumbai.

Achievements

- IITGN Overseas Research Experience Fellowship – Selected as a Visiting Research Scholar at the University of Birmingham, United Kingdom with Funding of 6800 USD by Indian Institute of Technology Gandhinagar, India [September 2024 to January 2025].
- Won the Best Poster Award and 100 Euros as cash price money at the University of Birmingham Research Showcase on 11-12-2026 for presenting findings on MOF bead-based water purification system.
- Journal Cover Page Recognition – Featured on the cover page of Environmental Science: Nano and Green chemistry research journals for outstanding research contributions.
- Link to ES Nano front cover page: https://pubs.rsc.org/en/article/11/6/2291/849954/Front-cover
- Link to Green chemistry back cover: https://pubs.rsc.org/gc/article/27/15/4004/894627/Back-cover
- RSC CNN Paper of the week- DOI: https://doi.org/10.1039/D5EN00056D
- Editor's choice article ASCE Publications- DOI: https://doi.org/10.1061/JHTRBP.HZENG-1406
- Gold Medalist – Secured first position in the second year of engineering with a 9.83/10 CGPA.
- Qualified GATE 2021 (Mechanical Engineering) with 98.12 percentile, securing an All India Rank of 2,257 among 120,594 candidates.

Position of Responsibility

- Assisted in discipline outreach activities for the Materials Engineering, IIT Gandhinagar, India.
- Teaching Assistant for the undergraduate course "Engineering Graphics"

Technical Skills & Instrumentation Experience

- **Hands-on Experience:** Metal-organic frameworks**,** Polymer composites, and nanomaterial synthesis**,** Adsorption and separation studies, Gas conversion experiments, SEM, FTIR, UV-Vis Spectroscopy, DMA, UTM, ICP-OES, TGA and DSC, Contact Angle Goniometer, BET surface area analyzers, XRD and XPS etc.

Reference Persons

- **Prof. Abhijit Mishra (Indian Institute of Technology, Gandhinagar)- amishra@iitgn.ac.in**
- **Prof. Superb Misra (Indian Institute of Technology, Gandhinagar)- smisra@iitgn.ac.in**
- **Prof. Kaling Taki (Indian Institute of Technology, Guwahati)- kaling.taki@iitg.ac.in**
- **Dr. Swaroop Chakraborty (University of Birmingham)- s.chakraborty@bham.ac.uk**